

FREDRIK PETTIT

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Education

University of British Columbia

B.Sc. Computer Science (Artificial Intelligence) / GPA: 3.8

Expected Graduation: May 2028

Vancouver, BC

Work Experience

AI Engineer Intern

Apr 2026 – Present

AEM

Langford, BC

- Shipped an AI support agent that resolves **42%** of staff and customer IT tickets, launched against an **800+** ticket backlog
- Engineered a RAG pipeline across bilingual docs that rewrites each query with an LLM before hybrid vector and keyword search on Azure AI Search over OpenAI embeddings, returning grounded answers with citations and a **~2s** first token
- Fine-tuned a cross-encoder reranker in PyTorch on **1,400** resolved tickets, lifting retrieval recall@5 from **74% to 88%**
- Cut monthly budget categorization from **6+ hours to 5 minutes** with a LangGraph agent that fixes vendors and GL codes across **1000+** transactions via Claude API, validated by an evaluation harness against finance's hand-done records

Software Engineering Intern

Jan 2026 – Apr 2026

ELLUSTROS

The Blue Mountains, ON

- Developed a recommendation engine that ranks **100+** office chairs against each user's measurements through layered filtering and weighted scoring across **15** ergonomic attributes, deployed as a FastAPI service powering a React Native app
- Cut validation defects per sprint by **32%** across the flagship mobile app by writing **40+** Jest and React Testing Library regression tests covering user sign-in flows and AI-driven measurement capture, run in a CI/CD pipeline on each merge

Teaching Assistant | CPSC 310, CPSC 110

Aug 2025 – Present

UBC Department of Computer Science

Vancouver, BC

- Supported **900+** students in UBC's largest computer science course, achieving a 4.9/5 average student evaluation score
- Built **250+** randomized HTML/JavaScript questions and Python autograders for CPSC 310 - Intro to Software Engineering

Projects

YouTube Shorts AI Optimizer | *Python, FastAPI, AWS (Lambda, DynamoDB), OpenCV, XGBoost*

- Developed an end-to-end video optimization pipeline automating thumbnail selection, title, and description generation for YouTube Shorts, growing a personal channel to **38,000+** subscribers, **45,000,000+** views, and **1,800,000+** likes
- Trained four XGBoost regressors on **3000+** videos to score thumbnail, title, description, and hashtag candidates, predicting engagement rate at **0.043 MAE** from OpenCV/Rekognition thumbnail features and sentence-transformer text embeddings
- Orchestrated an asynchronous AWS Step Functions workflow that fans video-analysis and inference jobs across Lambda functions via SQS queues, processing **230+** videos end to end with Gemini frame selection and Claude text generation

GitIntent - UC Berkeley AI LLM Hackathon | *Python, TypeScript, FastAPI, Redis Stack, voyage-code-3*

- Built a system that lets AI coding agents learn from developers and fellow agents by recording every change with the prompt behind it, then recalling failed attempts before repeating them, lifting performance **8.3%** across **36** DeepSWE runs
- Designed the hybrid retrieval engine on Redis Stack, fusing HNSW vector search over 1024-dimensional voyage-code-3 embeddings with BM25 ranking through Reciprocal Rank Fusion, deduplicating snapshot matches to reach **~93%** recall@5
- Engineered a content-addressed hashing scheme identical on client and server, giving idempotent, deduplicated writes

Inventory & Sales Automation Platform | *Spring Boot, React, TypeScript, PostgreSQL, Python, FastAPI*

- Built and operate custom infrastructure for a personal cross-platform retail business on Grailed, Depop, and eBay, logging **60+** international orders and **\$6K+** in transactions through a Spring Boot API and React/TypeScript frontend
- Automated order logging by parsing Gmail sale confirmations into the order tracker, and built a Python/FastAPI Playwright service that reposts 15 listings weekly on a 30-day bump-then-reduce price cycle to maximize buyer visibility
- Improved dashboard query performance **9×** across 10+ aggregated reporting views by adding composite PostgreSQL indexes

Technical Skills

Languages & Databases: Java, Python, C++, C#, SQL, R, JavaScript/TypeScript, PostgreSQL, DynamoDB, Redis
AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, OpenCV, embeddings, hybrid retrieval, reranking, fine-tuning, RAG
LLM & Agents: LangGraph, LangChain, LangSmith, OpenAI API, Anthropic Claude API, Google Gemini API, evals
Frameworks: Spring Boot, FastAPI, React, React Native, Expo, JUnit, Jest, Playwright, Pydantic, pandas, NumPy
Cloud & DevOps: AWS (Lambda, Step Functions, SQS, Rekognition), Azure AI Search, Docker, Gradle, Git, CI/CD